

Scalable Low-Energy Molecular Conformer Generation with Quantum Mechanical Accuracy

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Abstract

Molecular geometry is crucial for biological activity and chemical reactivity; however, computational methods for generating 3D structures are limited by the vast scale of conformational space and the complexities of stereochemistry. Here we present an approach that combines an expansive dataset of molecular conformers with generative diffusion models to address this problem. We introduce ChEMBL3D, which contains over 250 million molecular geometries for 1.8 million drug-like compounds, optimized using AIMNet2 neural network potentials to a near-quantum mechanical accuracy with implicit solvent effects included. This dataset captures complex organic molecules in various protonation states and stereochemical configurations. We then developed LoQI, a stereochemistry-aware diffusion model that learns molecular geometry distributions directly from this data. Through graph augmentation, LoQI accurately generates molecular structures with targeted stereochemistry, representing a significant advance in modeling capabilities over previous generative methods. The model outperforms traditional approaches, achieving up to tenfold improvements in energy accuracy and effective recovery of optimal conformations. Benchmark tests on complex systems, including macrocycles and flexible molecules, as well

as validation with crystal structures, show LoQI can perform low energy conformer search efficiently. The model code and dataset are available at <https://github.com/isayevlab/LoQI>.

Keywords: Conformer generation, Diffusion models, Stereochemistry, Structure-based drug discovery

1 Introduction

Molecular conformation plays a fundamental role in determining the behavior and function of chemical compounds across diverse domains, from materials science to pharmacology. A prototypical example of the critical role that molecular conformation plays is in specificity of protein active site, where the three-dimensional (3D) geometry of a ligand must complement the shape and electrostatics of the binding pocket to enable favorable interactions, such as hydrogen bonds and hydrophobic contacts [1, 2]. Poorly fitting conformers are known to cause steric clashes or suboptimal binding, leading to a reduction in affinity and efficacy. Stereochemistry adds another layer of complexity, as many biological targets are chiral and can distinguish between stereoisomers, often resulting in significant differences in potency or toxicity. One exemplary case is thalidomide, where only one enantiomer is therapeutically safe.

Despite advances in computational methods, generating accurate low-energy conformers at thousands-to-millions scale remains prohibitively expensive, limiting their use in high-throughput settings and creating a persistent bottleneck in structure-based molecular design. The importance of conformational complexity can be explained in the context of potential energy surfaces (PESs). For a molecule with N atoms, the PES is a mapping between interaction energy and atomic spatial positions: $E = E(q_1, q_2, \dots, q_{3N-6})$, where q_i are the internal coordinates that describe the positions of the atomic nuclei. From this definition it is clear that the conformational space grows exponentially with molecular size, leading to a "multiple minima problem," [3] where the number of local minima on PES (i.e., unique equilibrium geometries) scales approximately as 3^n for n rotatable bonds. This exponential scaling presents severe computational challenges for accurate and efficient conformational search, constituting a fundamental obstacle often described as the curse of dimensionality. Current conformer generation methods often struggle with molecules containing > 10 rotatable bonds, and force-field-based approaches frequently fail to reproduce quantum-mechanical conformer rankings. Structural deviations of optimized geometries with the same initial state often exceeding 0.6 Å RMSD between different force fields further underscore their inconsistency [4].

Modern conformational analysis must account for both thermodynamic stability and kinetic accessibility. For example, studies have shown that many bioactive conformations lie close to the global energy minimum, emphasizing the importance of understanding the conformational landscape [5]. Reliable generation of low-energy diverse conformers anchored to realistic energy landscapes is therefore critical for

downstream applications such as docking, virtual screening, and binding-free energy estimation [6–9].

Classical approaches for conformer generation rely primarily on heuristic algorithms and force fields. Examples include single-conformer generators such as CORINA, as well as ensemble-based methods like ETKTG, OMEGA [6], Conf-Gen [10], Frog2 [11], and Multiconf-DOCK [12]. Despite their efficiency, these methods frequently struggle with molecules characterized by significant flexibility, including macrocycles and "beyond rule-of-five" chemotypes [13]. As a consequence, classical force-field-based methods often yield significant errors in energy rankings and incomplete coverage of conformational ensembles, particularly as molecular complexity increases [4, 14].

Recent advances in machine-learned interatomic potentials (MLIPs), such as AIMNet2 [15], ANI [16], MACE [17], and UMA [18] have shown great promise in addressing these limitations. MLIPs can offer near-density functional theory (DFT) accuracy at substantially reduced computational cost, thereby enabling rapid and reliable geometry optimization.

In parallel, deep generative modeling techniques have gained widespread adoption in computational chemistry [19]. These models have been mostly trained on the GEOM dataset [20], which was created with the CREST software [21] based on semi-empirical quantum methods. One of the pioneering generative models, Torsional Diffusion [22] introduced a diffusion-based framework that operates within the torsion space. Although this approach is conceptually capable of incorporating stereochemical constraints in principle, its open-source implementation shows limited performance and frequently fails to produce correct stereoisomers.

Other group of methods directly model atomic coordinates, the group including GeoDiff [23], Molecular Conformer Fields (MCF)[24], and Diffusion Molecular Transformer (DMT)[25]. AvgFlow [26] introduces SO(3)-Averaged Flow, which improves conformer-generation training by analytically averaging the flow field over all 3D rotations, eliminating the need for explicit alignment. However, all of these models completely disregard stereochemistry, making them poorly suited for chiral or stereochemically rich molecules.

Additionally, GFlowNets [27] has been proposed to sample molecular conformations directly from the Boltzmann distribution focusing on equilibrium sampling in torsion-angle space, which should naturally favors low-energy conformers [27]. While this approach demonstrates accurate equilibrium conformer generation and energy-based sampling, it currently requires separate training for each individual molecule, restricting its general applicability and scalability.

In this work, we combine the strengths of MLIP energy evaluation with AIMNet2 and advanced generative models to develop a robust and widely applicable workflow for conformer generation. Specifically, we introduce **ChEMBL3D**, an extensive dataset containing over 250 million AIMNet2-optimized conformers for approximately 1.8 million unique molecules covering both charged and neutral states within an implicit solvent environment. We present **LoQI**(**L**ow-energy **Q**M **I**nformed conformer generative model), a stereochemistry-aware, equivariant generative model trained end-to-end on ChEMBL3D. In extensive tests on molecules with 0–25 rotatable bonds, LoQI

outperformed conventional cheminformatics tools and generative models while explicitly encoding stereochemistry. It also handles challenging cases like macrocycles and produces low-energy structures with high computational efficiency. These advances highlight the broad utility of our method in computational chemistry applications ranging from general conformational analysis to structure-based drug design.

2 Methods

2.1 ChEMBL3D dataset

The ChEMBL3D dataset was curated from the ChEMBL v34 database [28]. Initial processing involved generating protomers in neutral and non-ionized states using OpenEye FixpKa, and subsequently merging these with the original structures. OpenEye's Flipper tool has been used to enumerate stereocenters with missing annotation. Reference topology structures, including hydrogens but without explicit 3D coordinates, were created with Open Babel [29]. The molecules were sorted by the number of atoms and the training conformers were generated using the OpenEye Omega classic [6], followed by structural optimization using the FIRE algorithm [30] with a variant of the AIMNet2 [15] neural network potential, trained to reproduce the DFT energies within a CPCM solvation model. Unlike previous datasets that rely on classical force fields or semi-empirical quantum computations, all conformations were optimized under implicit solvation conditions to near DFT accuracy using AIMNet2.

The conformer dataset initially comprised approximately 505 million conformers prior to post-processing. Downstream steps included topology validation, energy-based filtering, and deduplication, resulting in a final dataset of 252 million processed conformers for 2,417,498 unique structures (representing 1,786,517 distinct ChEMBL molecules). Our deduplication procedure is based on the relative changes in the distance matrix, following the assumption that only one enantiomer of each pair is retained, in accordance with the protocol described in the work of Pracht [31]. After this filtering, the dataset contains 187 million conformers within a 6 kcal/mol energy window, 94 million within 2.5 kcal/mol, and 38 million within 1 kcal/mol. ChEMBL3D is orders of magnitude larger than previous 3D conformer datasets (e.g., GEOM), and to our knowledge it is the first to incorporate both an implicit solvent model and exhaustive stereochemistry/protomer enumeration at this scale. This makes it a uniquely comprehensive benchmark for generative models.

ChEMBL3D should be regarded as a collection of unique, relaxed conformers for ChEMBL molecules rather than a thermodynamical ensemble (e.g., GEOM-Drugs). Since a lot of recent generative molecular models employ only a limited number of conformations per molecule during training [25, 26, 32–35], we believe that providing a large-scale dataset of conformers with near QM accuracy will benefit the research community and facilitate further advances in generative modeling.

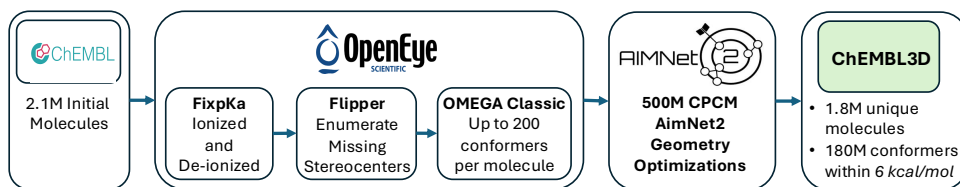


Fig. 1: Overview of the ChEMBL3Ddataset construction workflow. Molecules from ChEMBL v34 were expanded via protomer generation (OpenEye FixpKa), conformer generation (OpenEye Omega Classic), and stereoisomer enumeration (OpenEye Flipper). Conformers were geometry-optimized with AIMNet2 and filtered for broken topologies and duplicates. The final ChEMBL3Ddataset contains approximately 1.8 million unique molecules and 180 millions conformers within a 6 kcal/mol energy window.

2.2 Model Architecture and Training Objective

LoQI utilizes the recently introduced Megalodon architecture from [34], which shows great results for 3D molecule generation. This architecture combines a Diffusion Transformer (DiT) [36] layer, which processes invariant features, with an equivariant graph neural network (EGNN) [37] that updates atomic coordinates. Specifically, each transformer module integrates discrete atomic features, such as atom types and charges, and bond features, such as bond types and geometric features (pairwise distances and dot products). These combined features are processed through DiT layer which represents multi-head attention with adaptive layer normalization. After DiT layer, the resulting atom and bond features serve as additional inputs for coordinate updates performed by a structure layer based on EGNN. This structure-update step employs distance-based EGNN updates enhanced by a cross-product term refining atomic coordinates.

In adapting the Megalodon framework, our approach simplifies the prediction task by exclusively focusing on atomic coordinates, using atom types, charges, and bonds solely as inputs. This design choice capitalizes on Megalodon’s demonstrated strengths in accurately predicting molecular coordinates, thus streamlining computational efficiency without explicit modeling of discrete bond or atom-type outputs.

Our training procedure employs a diffusion-based approach consistent with few recent developments in generative modeling [32, 33]. Specifically, we utilize weighted cosine noise schedules within a standard denoising diffusion probabilistic model (DDPM). Comprehensive technical details and exact model configurations can be found in the original Megalodon paper [34] and implementation. Remarkably, we observe that for conformer generative task, only 25 diffusion steps are sufficient to reach peak performance.

3 Results

The LoQI model has been trained to generate lowest energy conformer from ChEMBL3D utilizing a diffusion training objective. Rather than navigating the space

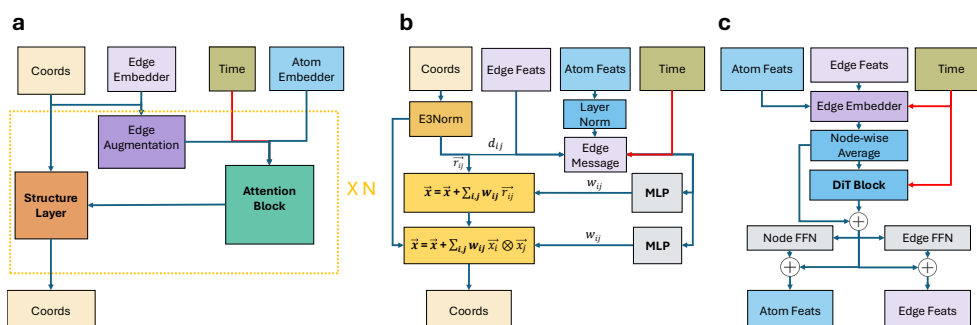


Fig. 2: LoQI is a Megalodon-based adaptation [34] tailored for denoising diffusion over 25 steps, where the model iteratively predicts denoised atomic coordinates from progressively noised inputs. (a) The overall neural network architecture processes discrete atom features, bond features, diffusion timestep embeddings, and atomic coordinates; (b) A structure layer based on EGNN performs equivariant coordinate updates using equivariant and invariant features; (c) A DiT-style transformer block updates invariant atom and edge features conditioned on diffusion timestep.

with predefined rules, LoQI learns directly to produce chemically valid, low-energy 3D geometries. This data-driven modeling strategy is not restricted by predefined rotatable bonds or fragment libraries, which crucially allows our model to efficiently handle the generation of complex molecular geometries. In the following sections, we benchmark LoQI against classical and generative baselines on multiple datasets to assess its performance and generalizability.

3.1 Hold-Out Set Performance

The generalization of LoQI has been rigorously evaluated using a newly constructed hold-out set of 37,075 molecules (including diverse isomeric and protonation states) sampled from ChEMBL3D. None of these molecules were used during training, allowing us to assess performance in a realistic setting, where the model must generate low-energy conformers without relying on memorized patterns or specific heuristics. The results are presented in Table 1.

LoQI consistently outperforms all baselines across key metrics. It achieves a median relative energy of 0.33 kcal/mol, significantly lower than both retrained generative baselines (e.g., 0.45 kcal/mol for DMT-B) and classical methods like RDKit and Omega (3.39 and 1.42 kcal/mol, respectively). Notably, LoQI finds conformers within 0.1 kcal/mol of the minimum of the dataset in 38.5% of molecules, nearly doubling all other methods, without pretraining on our dataset. In other words, generating and optimizing a single conformer using LoQI achieves, in 38.5% cases, results equivalent to optimizing up to 200 conformers produced by OpenEye Omega Classic. Furthermore, in 14.0% of the cases, the generated conformer has a lower energy compared to the reference conformer in ChEMBL3D, highlighting the ability of the model to perform low-energy conformer search.

The generative model has been trained on the AIMNet2 optimized conformers; therefore, it is worth confirming that the conformers with lower energy than those in

the dataset are not an artifact or errors of the MLIP model. We selected 88 examples from the test set in which the model, using single-shot generation, identified conformers at least 1 kcal/mol lower in energy compared to the dataset reference. To validate these results, we performed density functional theory (DFT) calculations with a CPCM implicit solvent model. Of the 77 molecules that successfully completed DFT geometry optimization, only one had a higher energy (0.8 kcal/mol) compared to the reference. For 56 molecules, DFT optimization confirmed an energy improvement of at least 0.5 kcal/mol , validating the accuracy of our generative model (see Figure 3 for examples and plot). These results demonstrate that LoQI can efficiently search conformers with DFT-level accuracy at computational costs significantly lower than direct DFT-level geometry optimization.

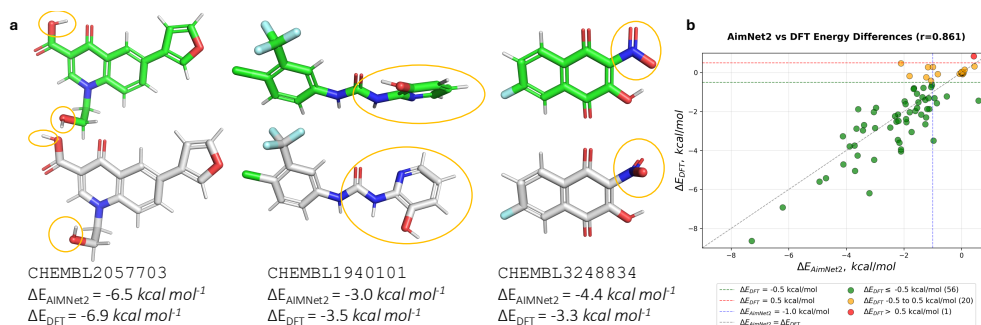


Fig. 3: Comparison of AIMNet2 and DFT energies for molecules where LoQI identified a lower-energy conformer via single-shot generation. (a) Three representative examples illustrating structural differences between the ChEMBL3D reference conformers (gray) and the generated conformers (green), along with corresponding energy differences computed using AIMNet2 and DFT. (b) Correlation between energy differences computed using AIMNet2 and those obtained from DFT, both evaluated on DFT-optimized structures across all 77 successfully optimized molecules. The strong agreement highlights the reliability of AIMNet2 for accurately guiding low-energy conformer generation.

LoQI excels not only on conformational energy metrics but also structurally. Geometry optimization converged of the generated conformers converged in nearly every case, perserving the molecular topology defined by the input structure. Additionally, the median relaxation energy (ΔE_{relax})—the energy adjustment required after post-generation geometry optimization—is minimal (3.48 kcal/mol), indicating that LoQI produces conformations remarkably close to true minima, requiring little subsequent refinement. This reduces the need for costly refinement steps, making the approach highly suitable for applications requiring high-throughput, structure-based analysis.

Classical methods, while highly reliable in preserving topology (e.g., RDKit at 99.5%), struggle to generate accurate low-energy conformations. RDKit finds near-optimal conformers in only 10.4% of molecules and improves upon the dataset

minimum in just 3.1%. Omega performs similarly, reflecting the limitations of heuristic-driven sampling in covering the true conformational landscape.

Among the generative model baselines, retrained DMT-B is the most competitive, achieving 33.4% within 0.1 kcal/mol of the minimum. That model exhibits a substantially higher relaxation energy (43.42 kcal/mol), indicating its generated conformers substantially deviate from true local minima and revealing a fundamental limitation in learning the underlying energy landscape. Models without retraining perform substantially worse, underscoring the value of both the method and the training data.

Together, these results support a broader conclusion: LoQI represents a practical solution to the long-standing problem of efficient low-energy conformer search without relying on heuristics, predefined rotatable bonds, or fragment templates.

Table 1: Performance comparison of low-energy conformer generation methods on the ChEMBL3D hold-out set. Metrics include optimization success, topological fidelity, and energy-based accuracy.

Method	Median $\Delta E_{\text{relative}}^{\text{opt}}$ ¹ (kcal/mol)	% $\Delta E_{\text{relative}}^{\text{opt}}$ ¹ < 0.1 kcal/mol	% $\Delta E_{\text{relative}}^{\text{opt}}$ ¹ < -0.1 kcal/mol	Minimization Converged, %	Topology Preserved, %	Median ΔE_{relax} ² (kcal/mol)
RDKit	3.39	10.4	3.1	99.5	99.5	64.50
OEOmega	1.42	15.7	13.0	99.8	99.8	13.19
TorsDiff	2.62	15.3	7.7	99.0	97.4	63.09
DMT-B	1.81	19.9	11.5	99.8	99.8	82.70
DMT-L	1.72	20.8	12.4	99.9	99.8	86.43
TorsDiff ³	4.66	7.7	2.7	94.6	87.3	92.02
DMT-B ³	0.45	33.4	12.5	99.5	95.7	43.42
LoQI	0.33	38.5	14.0	100	99.9	3.48

¹ $\Delta E_{\text{relative}}^{\text{opt}}$ is the energy difference between the optimized conformer and the dataset’s lowest-energy conformer for the same molecule.

² ΔE_{relax} is the energy difference between the generated and optimized conformer.

³Indicates models retrained using the same protocol as LoQI on ChEMBL3D.

3.2 LoQI is a Stereochemistry-Aware Model

A significant limitation of current all-atom generative models operating directly on molecular graphs is the lack of explicit stereochemical encoding. Unlike torsion-based methods such as Torsional Diffusion, which theoretically preserve stereochemistry by modeling conformers in torsional space, Cartesian coordinate-based models rely only on relative atomic arrangements and element types. As a result, they are unable to recover stereochemical configurations such as E/Z isomerism and R/S chirality, which makes these tools less practical.

We address this challenge by introducing a simple yet effective graph augmentation that encodes stereochemical information directly into the input graph (see Figure 4). For E/Z stereochemistry, we add four auxiliary edges between neighbors of

each double bond with possible E/Z isomerism: two “joint” Z-bonds connecting adjacent substituents and two “diagonal” E-bonds connecting across the double bond. This distinguishes stereoisomers based on relative substituent positioning. For R/S stereocenters, the lowest-priority (according to Cahn-Ingold-Prelog rules) atom connects to the remaining three via undirected edges, while the other three form a directed cycle consistent with the stereochemical configuration. These augmentations, combined with directional message passing, provide sufficient context for the neural networks to drive stereochemistry-aware 3D structure generation.

The stereochemical augmentation is both architecture-agnostic and easily generalizable to other stereochemical types, such as axial or planar chirality. It effectively bridges a gap between the classical cheminformatics tools that inherently handle stereochemistry and modern generative approaches.

Table 2 demonstrates the effectiveness of this strategy. LoQI is capable to preserve defined stereochemistry, with 0.956 R/S and 0.992 E/Z accuracy, on par with traditional methods like RDKit and OpenEye Omega, and substantially outperforming all previous generative models. Notably, even retrained diffusion models such as Torsional Diffusion struggle with E/Z stereochemistry, despite our best efforts to enforce correct initialization and labeling. This suggests that torsional generative models, while inherently more structured, still lack reliable mechanisms for the enforcement of stereochemistry during sampling.

The proposed stereochemical graph encoding approach not only enhances LoQI’s practical applicability but also represents, to our knowledge, the first successful integration of explicit stereochemical encoding into an all-atom diffusion-based generative model. This advancement significantly expands the scope of generative models for reliable and accurate application across diverse chemical and pharmaceutical use cases.

3.3 Impact of Rotatable Bonds

A critical challenge in molecular conformer generation is the exponential complexity introduced by increasing number of rotatable bonds. To specifically assess performance under this challenge, we created an additional test subset, which we refer to as ChEMBL3D-XL. This subset consists of approximately 300 molecules per number of rotatable bonds selected to span up to 25. For each molecule, we generated up to 100 conformers using LoQI, RDKit ETKDG, and OE Omega Classic, followed by optimization with the same neural network potential used during ChEMBL3D creation. We then evaluated the probability of recovering the reference minimum conformer as a function of the number of trials. We define success for this performance assessment as at least one attempt producing a conformer within within 0.1 kcal/mol of the value in the dataset. We also look at the fraction of molecules for which we were able to find better conformation with 0.5 kcal/mol margin.

The results highlight the significant gain in performance of the LoQI model, especially as molecular complexity increases with the number of rotatable bonds. As shown on Figure 5, the model’s accuracy improves with increasing numbers of rotatable bonds leading to an order of magnitude higher success rate for molecules with large number of rotatable bonds compared to OE Omega and RDKit.

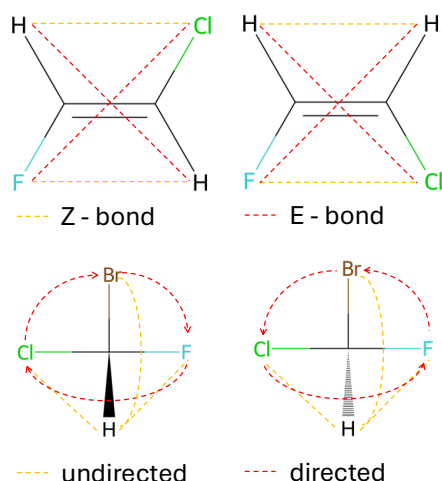


Fig. 4: Stereochemistry-aware graph augmentations. **Top:** For E/Z, four auxiliary bonds are added—two *Z*-bonds (joint) and two *E*-bonds (diagonal)—between neighbors of a double bond. **Bottom:** For R/S, the lowest-priority atom connects to the others via undirected edges; the rest form a directed cycle reflecting stereochemistry.

Table 2: Stereochemistry accuracy of conformer generation methods on the ChEMBL3D hold-out set. Reported values indicate the fraction of molecules with correctly generated configurations for R/S and E/Z stereocenters.

Method	R/S Accuracy ¹	E/Z Accuracy ²
RDKit	0.993	0.993
OEOmega	0.993	0.985
TorsDiff	0.993	0.734
DMT-B	0.377	0.706
DMT-L	0.385	0.740
TorsDiff ³	0.993	0.502
DMT-B ³	0.793	0.990
LoQI	0.956	0.992

¹ R/S accuracy: fraction of molecules with correctly reproduced absolute stereochemistry at tetrahedral centers.

² E/Z accuracy: fraction of molecules with correct stereochemistry at double bonds.

³ Indicates models retrained using the same protocol as LoQI on ChEMBL3D.

3.4 Fast and Accurate Estimation of Energy Differences Between Bound and Unbound Molecular States

To demonstrate the utility of LoQI in estimating energy differences between ligand-bound and unbound states, we evaluated it on the Platinum Diverse dataset [5], a widely used benchmark containing 2,879 ligand crystal structures. Although this dataset was originally proposed for conformer generation benchmarking, it consists of crystal-bound conformers that often deviate significantly from the lowest-energy gas-phase or solution-phase geometries due to intermolecular interactions and crystal packing effects [5, 38]. Since LoQI is trained to predict low-energy conformers of isolated molecules in CPCM implicit solvent, we do not report standard conformer generation metrics on this set (see Appendix A for further discussion). Instead, we use the energy differences between experimentally observed bound conformers and computationally predicted low-energy structures as a rigorous test of the model performance.

For each ligand, we generated conformational ensembles comprising 500 structures using LoQI and RDKit’s ETKDG algorithm. We identified the lowest-energy conformer from each ensemble as the reference minimum. The observed energy differences showed a median of 1.84, kcal, mol⁻¹, consistent with the typical magnitude of

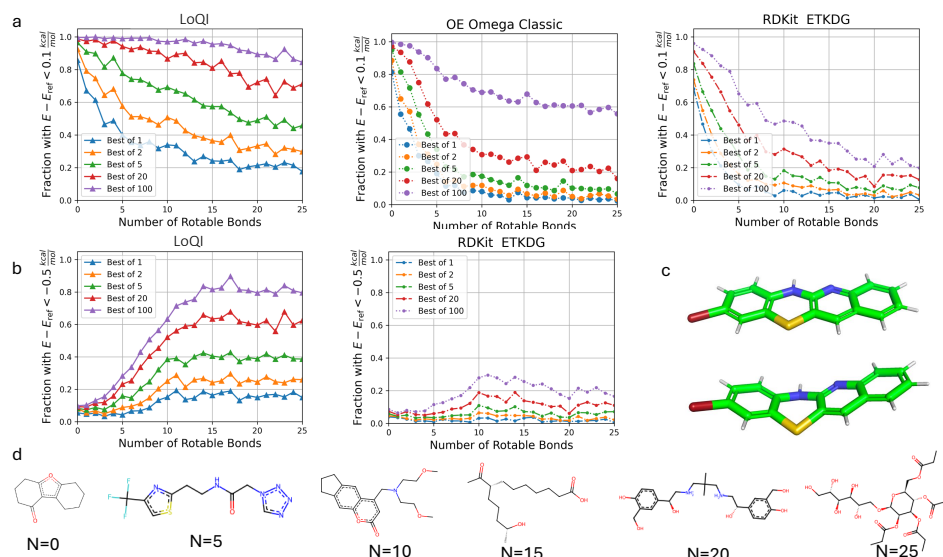


Fig. 5: (a) Fraction of molecules for which the LoQI finds the minimum-energy conformer within a specified number of trials (1, 2, 5, 20, 100), as a function of the number of rotatable bonds. (b) Fraction of molecules where LoQI identifies a conformer lower in energy than the ChEMBL3D dataset reference conformer by at least 0.5 kcal/mol. (c) Example of a molecule with zero rotatable bonds for which LoQI finds a lower-energy conformer than those in ChEMBL3D. (d) Representative molecules with different numbers of rotatable bonds, as defined in RDKit.

the change in energy from the crystal packing (Figure 6a). Several structures exhibited large deviations (> 10 , kcal, mol⁻¹), reflecting substantial structural differences between the crystal-bound and isolated conformational preferences.

Figure 6b illustrates the accuracy tradeoff between the number of sampled conformers and the estimated energy differences between crystal-bound and isolated minima. Notably, chemical accuracy (mean absolute error < 1 , kcal, mol⁻¹) is achieved by LoQI with as few as five sampled conformers, whereas RDKit's ETKDG fails to reach this threshold even after generating 100 conformers. Quantitatively, expanding the conformer ensemble from one to 99 conformers reduces mean absolute prediction error from 2.29 to 0.14, kcal, mol⁻¹ for LoQI, compared to a reduction from 5.24 to 1.31, kcal, mol⁻¹ for RDKit ETKDG. This contrast highlights the capability of generative models to rapidly and accurately capture nuanced conformational energy landscapes that would be difficult to achieve with the rule-based heuristics of traditional methods.

3.5 Performance on Macrocycles and Complex Stereoisomers

Macrocyclic molecules are a particularly demanding class of chemical compounds for conformer generation due to their large ring structures, inherent flexibility, and complex conformational landscapes. We selected six representative macrocycles from the

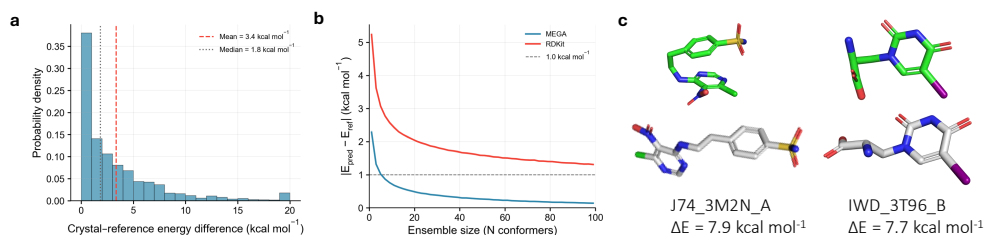


Fig. 6: Neural-network-driven estimation of conformational energy differences on the Platinum Diverse dataset. (a) Distribution of energy differences between crystal-bound and isolated minimal-energy conformers, highlighting significant energy deviations due to crystal packing constraints. (b) Mean absolute error (MAE) in energy difference estimation versus conformer ensemble size. LoQI rapidly achieves chemical accuracy compared to RDKit's ETKDG. (c) Representative crystal structures demonstrating geometric deviations between crystal-bound (grey) and isolated minimal-energy conformations (green). Crystal-bound structures are consistently more open and strained compared to minima.

MPCONF196 benchmark dataset [39] to serve as a targeted assessment of LoQI's accuracy and transferability. For each macrocyclic compound, ensembles of 100 conformers were generated, and the lowest-energy conformer most closely resembling the high-level quantum-chemical reference structures (CCSD or MP2 optimized) was identified.

The results, illustrated in Figure 7, demonstrate that LoQI consistently generates accurate macrocyclic conformers closely aligned with reference structures, achieving root-mean-square deviation (RMSD) ranging from approximately 1.0 Å (for smaller macrocycles such as POXTR) to about 1.9 Å (for larger, more flexible cases like SANGLI).

Additionally, we evaluated the performance of our method on taxol, a molecule recognized for its challenging stereochemistry and structural complexity. Figure 7b compares the generated taxol conformer (in green) against its experimental crystal structure obtained from the Cambridge Structural Database (CSD) (in grey). The close alignment, 1.2 Å, underscores our model's robustness in accurately reproducing intricate stereochemical details. These results collectively highlight LoQI's generalizability to complex molecules with numerous stereocenters.

4 Conclusions

In this study, we introduced ChEMBL3D, the largest publicly available dataset of high-quality, DFT-level molecular conformations, comprising over 250 million optimized conformers for 1.8 million molecules with diverse protonation states and stereochemical configurations. Leveraging this dataset, we trained LoQI, a stereochemistry-aware diffusion model tailored explicitly for accurate and computationally efficient low-energy conformer generation. LoQI represents the first generative model trained on complex

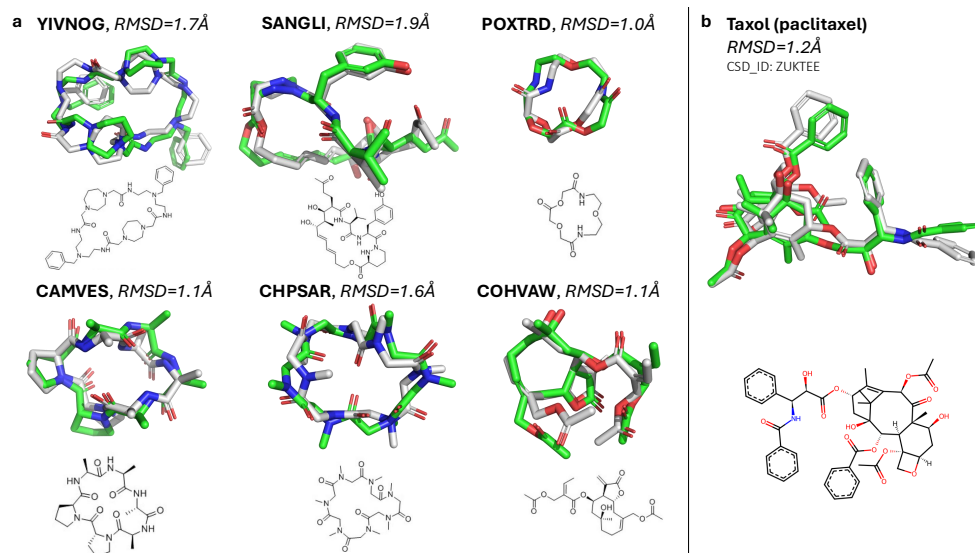


Fig. 7: Performance on challenging macrocycles and complex stereoisomers.

(a) Examples of macrocyclic conformers generated for selected molecules from the MPCONF196 dataset. Generated conformers (green) are overlaid with their reference structures (grey). RMSD values highlight strong structural agreement. (b) Example of taxol, a molecule known for challenging stereochemistry, comparing generated conformer (green) with its experimental crystal structure from CSD (grey).

molecules sourced from ChEMBL. Critically, ground truth conformers are optimized under realistic conditions with implicit solvent.

The LoQI model overcomes the challenge of conformational sampling and the need for exhaustive multi-conformer searches by directly generating low-energy 3D geometries in a single shot, bypassing the heuristic-driven exploration of vast conformational space. This enables it to consistently produce conformers within 0.1 kcal/mol of the minimum-energy structure for a large fraction of molecules while requiring minimal post-generation optimization. As a result, LoQI effectively integrates the conformational search and low-energy selection steps into a single generative process, eliminating the computational cost of enumerating and optimizing large conformer ensembles.

Through extensive benchmarks and rigorous evaluations, LoQI significantly surpassed traditional heuristic-driven algorithms (RDKit, Omega) and contemporary deep generative models (Torsional Diffusion, Diffusion Molecular Transformer) in both accuracy and efficiency, establishing new standards in conformer generation performance. Key innovations of LoQI include its effective handling of complex stereochemistry via a graph augmentation strategy, and superior geometric quality. Furthermore, its notable performance on complex macrocycles and challenging molecules highlights its broad applicability across key application areas where molecular conformations dictate chemical properties.

Despite its significant progress, our approach has limitations that should be acknowledged. First, the LoQI model is trained on implicitly solvated conformers optimized at the DFT level; therefore, explicit solvent effects, atomically resolved biomolecular environments, and QM correlation effects beyond DFT with CPCM are not captured. Second, while the model exhibits excellent generalization across various chemotypes, molecules that are far outside the chemical diversity of the ChEMBL dataset may not be accurately represented. However, such cases are expected to be training data limited and not errors related the generative model design, i.e., they can be addressed in future expanded training efforts. Lastly, although computational cost and scalability have been greatly improved compared to existing methods, they could still pose practical challenges in ultra-large virtual screening campaigns or related scenarios.

Looking forward, LoQI supports a generative modeling paradigm shift with substantial potential to accelerate computational workflows in pharmaceutical discovery, virtual screening, and molecular design. Future developments could extend LoQI to explicitly incorporate biomolecular environments, metal-organic structures, and further QM accuracy enhancements that bridge the gap between theoretical models and practical applications in chemistry, biology, and materials science.

To facilitate widespread adoption, reproducibility, and community-driven advancements, we openly release both the ChEMBL3D dataset and the LoQI model. The model's code, references to datasets, and other relevant resources are available at <https://github.com/isayevlab/LoQI>. We believe this resource will significantly benefit researchers and practitioners across various fields, inspiring new methodological developments and fostering deeper collaboration within the computational chemistry and artificial intelligence communities.

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Appendix A Platinum-Diverse Conformer Search Benchmark

The Platinum Diverse dataset, comprising 2,879 experimentally determined ligand crystal structures [5], is a widely accepted benchmark for evaluating small-molecule conformer generation methods. However, crystal-packing effects and protein-ligand interactions frequently cause bound ligand conformers to deviate substantially from their gas-phase or implicit-solvent energy minima. Our model, trained exclusively on *in silico* conformers—specifically, ChEMBL molecules optimized using the CPCM–AIMNet2 neural network potential to approximate implicit-solvent DFT conditions—is inherently biased toward lower-energy, solution-phase geometries. As such, we anticipated partial but not complete transferability to the more energetically diverse and structurally constrained conformations found in the Platinum Diverse dataset.

Despite these inherent limitations, LoQI consistently yields the lowest mean and median RMSD values across all ensemble sizes evaluated (Table A1). Notably, the model achieves a median RMSD of 0.45 Å at an ensemble size of $N = 250$, surpassing RDKit (0.52 Å) and ETKDG (0.54 Å), and matching performance of computationally intensive genetic-algorithm methods.

These findings demonstrate that a generative model trained exclusively on low-energy solution-phase conformations effectively generalizes to the distinct conformational landscape of protein-bound ligands. More broadly, this benchmark underscores the potential of data-driven 3D generative models to overcome the intrinsic complexity of conformer searches, even under conditions significantly divergent from the training distribution.

Table A1: Arithmetic mean and median RMSD (Å) for the Platinum Diverse dataset. Values in **bold** indicate the best (lowest) performance in each column; values statistically indistinguishable from the best (within ± 0.02 Å) are underlined. Bold and underlined entries represent the LoQI performance when at least one other method is within this tolerance range.

Method	10		50		250	
	Mean	Median	Mean	Median	Mean	Median
Balloon DG	1.10	0.97	1.00	0.86	0.92	0.77
Balloon GA	1.22	1.10	0.90	0.80	0.72	0.63
RDKit	1.00	<u>0.89</u>	0.77	<u>0.64</u>	0.63	0.52
ETKDG	0.98	<u>0.87</u>	0.77	<u>0.66</u>	0.63	0.54
Frog2	1.18	1.19	0.93	0.85	0.75	0.65
Multiconf-DOCK	0.99	<u>0.89</u>	0.84	0.72	0.80	0.69
LoQI	0.95 $\pm$ 0.01	<u>0.88</u>	0.74 $\pm$ 0.01	<u>0.65</u>	0.58 $\pm$ 0.01	0.45